

Multiple Choice

1. **Answer:** A
Options: A) $10/e$; B) 10; C) $10 e$; D) e^{10}
Note: Correct option: A - $10/e$
2. **Answer:** B
Options: A) 4; B) 6; C) 8; D) 9
Note: Correct option: B - 6
3. **Answer:** A
Options: A) $y = 2x + 1$; B) $y = 2x$; C) $y = x + 2$; D) $y = x + 1$
Note: Correct option: A - $y = 2x + 1$
4. **Answer:** C
Options: A) 4; B) 8; C) 12; D) 6
Note: Correct option: C - 12
5. **Answer:** B
Options: A) 0; B) 2; C) 1; D) 3
Note: Correct option: B - 2

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, True'.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** We can write $f(x) = \sqrt{5x^2 + 2x\sqrt{5} + 1} + x\sqrt{5}$
 $= \sqrt{(x\sqrt{5} + 1)^2} + x\sqrt{5}$
 $= |x\sqrt{5} + 1| + x\sqrt{5}$. If $x \leq -\frac{1}{\sqrt{5}}$, then

$$f(x) = |x\sqrt{5} + 1| + x\sqrt{5} = -x\sqrt{5} - 1 + x\sqrt{5} = -1.$$

If $x \geq -\frac{1}{\sqrt{5}}$, then $f(x) = |x\sqrt{5} + 1| + x\sqrt{5}$

$$= x\sqrt{5} + 1 + x\sqrt{5}$$

$$= (x\sqrt{5} + 1) + (x\sqrt{5} + 1) - 1$$

≥ -1 . Thus, the minimum value of $f(x)$ is -1 .

Note: Students should show all steps in their mathematical solution.

12. **Answer:** By the Integer Root Theorem, the possible integer roots are all the divisors of 6 (including negative divisors), so they are $-6, -3, -2, -1, 1, 2, 3, 6$.

Note: Students should show all steps in their mathematical solution.

End of Answer Key